

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Secchi Disk	WQP 005	1.1	

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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC
1.1	9/1/23	Included instructions for secchi depth > site depth	TAC

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1. Title

Standard Operating Procedure for measuring seawater transparency using a Secchi Disk.

2. Applicability

This SOP applies to determining changes in seawater transparency at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to measure coral reef seawater transparency using a Secchi Disk at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and other parties interested in coral reef seawater transparency data following the Quality Assurance Project Plan.

4. Method

Secchi Disk Method in US EPA LG-402 (2013)

5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.

6. General Precautions

- 6.1. Pay attention to changes in the state of the rope attached to the Secchi Disk, it may shrink or twist with use and affect the accuracy of the depth measurement.
 - 6.1.1. Change as needed
- 6.2. Always maintain two points of contact when deploying the Secchi disk on the boat to avoid losing the instrument overboard.
- 6.3. Ensure the boat engines are idle and the captain approves of the Secchi disk being deployed overboard before deploying the Secchi depth to ensure that the line does not become entangled with the boat propeller.

7. Training

- 7.1. Personnel shall be trained in the proper use and reading of a Secchi Disk.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Secchi Disk
- 8.2. At least 45m rope marked at 0.5 m intervals or less.
- 8.3. Field Sampling Log Form (Appendix F)

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9. Procedure

Seawater depth transparency measurements will be obtained using a 20 cm diameter black and white Secchi disk with a non-stretchable rope marked at 0.5 m intervals.

- 9.1. Lower the Secchi disk in a slow and steady mode until it is no longer visible to the observer.
 - 9.1.1. Lower the Secchi disk from the shady side of the boat.
 - 9.1.2. Reading should preferably be performed on clear-sky days.
 - 9.1.3. Minimize glare by looking as close as possible to the water.
 - 9.1.4. Do not use sunglasses or hats while reading the Secchi disk.
- 9.2. Slowly raise the disk until it just becomes visible once again.
- 9.3. Record the reading in Field Form (Appendix F).
 - 9.3.1. The depth is quantified to the nearest 0.25 m interpolated from the 0.5 m depth intervals of the Secchi disk line.
 - 9.3.2. If the secchi depth is greater than the site depth, the secchi depth should be recorded as "> X m" where X is the depth of the site.
 - 9.3.3. In case any of the measurements differ by more than 0.5 m, the measurement shall be repeated.
- 9.4. Repeat the process two more times for a total of three times.
- 9.5. Pull the Secchi disk back up into the boat.

10. Documentation

- 10.1. Field data will be recorded in Field Form (Appendix F).
- 10.2. Field forms shall be kept in the designated file cabinet at the CCRI office.
- 10.3. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.4. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.5. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. After each sampling event, rinse the Secchi disk and the rope thoroughly with fresh water.
 - 11.1.1. Allow the rope to air dry prior to storage.

12. Troubleshooting

- 12.1. If any of the measurements differ by more than 0.5 m the measurement should be repeated.
- 12.2. Two standardized, identical Secchi disks will be on hand at CCRI if one is lost or irreparably damaged to maintain consistency across samples.

13. QA/QC

- 13.1. As per QAPP Table 4, seawater transparency should be measured with a Secchi disk in meter (m) with an accuracy of $\pm 20\%$ and a precision of $< 20\%$.
- 13.2. It is best to make Secchi depth measurements between 10:00 am and 2:00 pm.

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- 13.3. The depth at which the Secchi disk disappears and reappears is quantified 3 times to the nearest 0.25 m interpolated from the 0.5 m depth intervals of the Secchi disk line.
- 13.4. If any of the measurements differ by more than 0.5 m the measurement shall be repeated.

14. References

- 14.1. U.S. Environmental Protection Agency, Office Of Research And Development. 1986.
- 14.2. U.S. Environmental Protection Agency, LG-402, Revision 11, March 2013. Standard Operating Procedure for Zooplankton Sample Collection and Preservation and Secchi Depth Measurement Field Procedures